

The Pauli Exclusion Principle

John W Daily

Identical Particles

- Consider the wave equation for two identical particles in a box

$$-\frac{\hbar^2}{2m} \left(\frac{\partial^2 \psi}{\partial x_1^2} + \frac{\partial^2 \psi}{\partial x_2^2} \right) + V\psi = 0$$

- Where

$$\psi = \psi(x_1, x_2)$$

$$V = V(x_1, x_2)$$

- If the particles are interchanged, the probability density of the system should be unchanged

$$p(1,2) = p(2,1)$$

Symmetry Requirement

- For this to be the case, we must have

$$\psi(1,2) = \psi(2,1) \quad \text{Symmetric}$$

$$\psi(1,2) = -\psi(2,1) \quad \text{Anti-symmetric}$$

- since

$$p(\vec{r},t)dV = \Psi^*(\vec{r},t)\Psi(\vec{r},t)dV = |\Psi(\vec{r},t)|^2$$

Classifying

- Particles with half-integer spin quantum number have anti-symmetric wave functions
 - **Fermions**
 - Electrons, protons, neutrons
 - Atoms and molecules with an odd number of electrons, protons and neutrons
- Particles with integer spin quantum number have symmetric wave functions
 - **Bosons**
 - Photons
 - Atoms and molecules with an even number of electrons, protons and neutrons
- But this does not prove the exclusion principal.

Leading to the Exclusion Principle

- Assume we can write

$$V = V(x_1) + V(x_2) = V(1) + V(2)$$

- Then

$$\psi_{\alpha,\beta}(1,2) = \psi_{\alpha}(1)\psi_{\beta}(2)$$

- Where α and β denote the quantum state of each particle

- We can now construct a wave function that satisfies the symmetry requirement

$$\psi_{s,a}(1,2) = \frac{1}{\sqrt{2}} \left\{ \psi_{\alpha}(1)\psi_{\beta}(2) \pm \psi_{\alpha}(2)\psi_{\beta}(1) \right\}$$

- where s and a stand for symmetric and antisymmetric.
- This clearly satisfies $\psi(1,2) = \pm\psi(2,1)$
- Note, however, that if the two particles are in the same quantum state, and the wave function is antisymmetric, then

$$\psi \rightarrow 0$$
- Thus **fermions** cannot occupy the same quantum state at the same time!

Pauli Exclusion Principle

- No two electrons of a multi-electron atom can assume the same quantum state.

The structure of all multi-electron atoms derives from this fact.